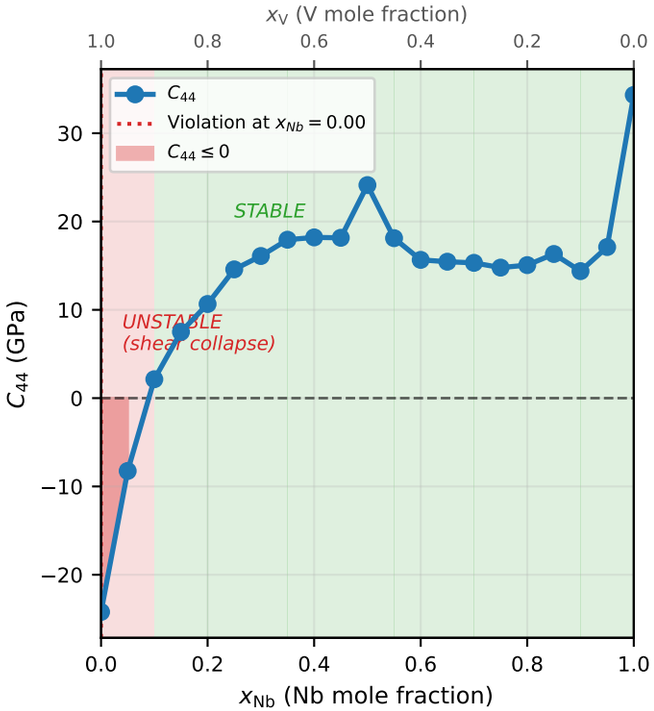
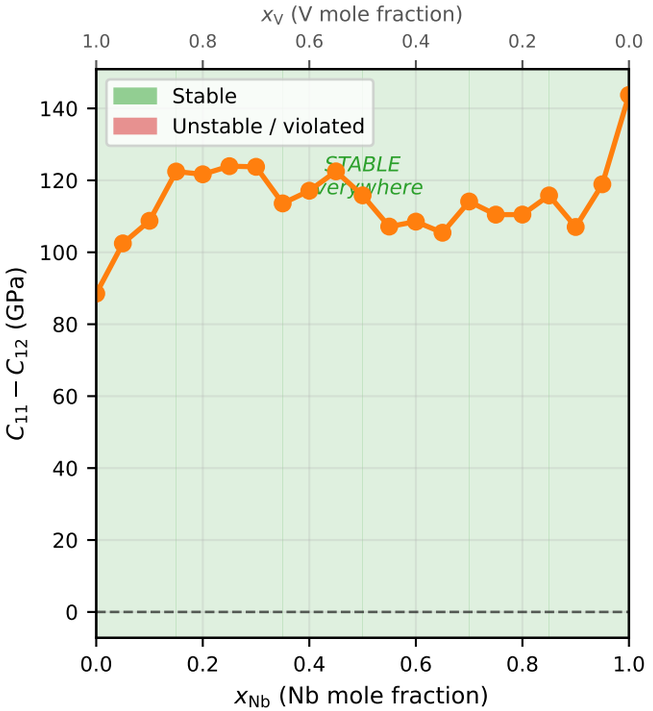
Nb-V BCC — Born Mechanical Stability Criteria (SQS/DFT, PhaseForge v2)
Only $C_{44} > 0$ is violated, for $x_V \gtrsim 0.90$ ($x_{Nb} \lesssim 0.10$). Use $x_{Nb} = 0.10$ as safe PFM precipitate endpoint.

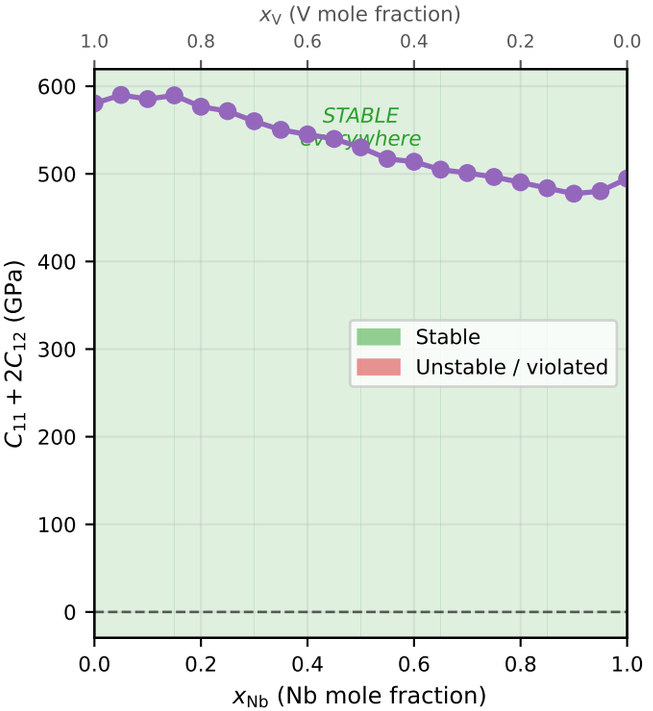
Criterion 1: $C_{44} > 0$ (shear stability on {100})



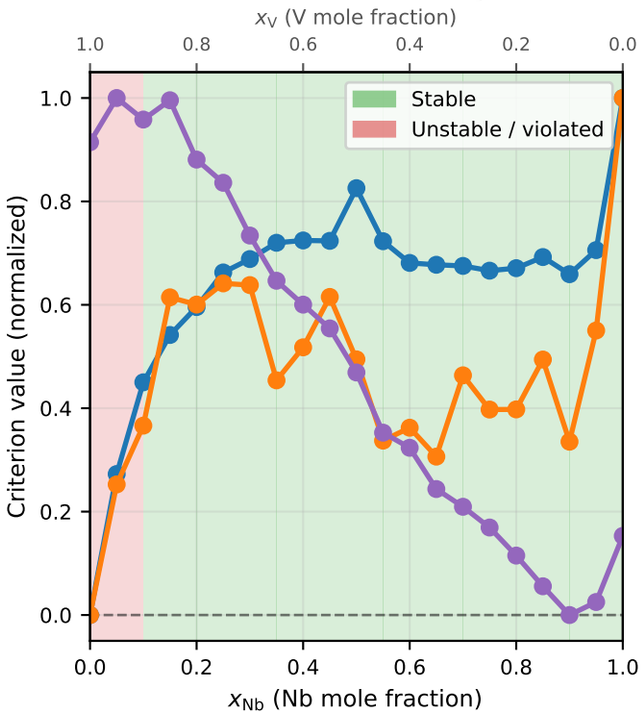
Criterion 2: $C_{11} - C_{12} > 0$ (tetragonal shear stability)



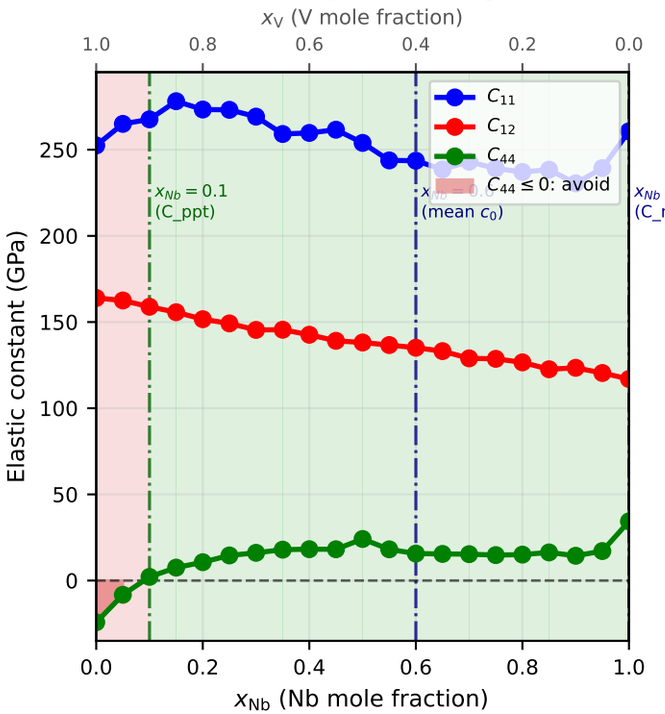
Criterion 3: $C_{11} + 2C_{12} > 0$ (bulk / hydrostatic stability)



All Born criteria (combined stability)



Stiffness constants with stability and PFM points



Born Stability Map (red = violated, green = satisfied)

